Objectives:

(i) To understand different types of earthing systems.

(ii) To analyze the layout and requirements of telecommunication wiring within buildings.

(iii) To examine procedures for inspection and testing.

(iv) To familiarize with the major national and international electrical standards.

(v) To explore various lightning protection techniques

(vi) To evaluate design examples provided for practical understanding.

Earthing Systems:

Earthing provides a safe, low-resistance path for fault currents to flow into earth, preventing shocks and fire hazards.

There are 3 major types of earthing:

(i) Copper Rod Earthing: A long copper rod is driven deep into the soil to achieve low earth resistance and stable grounding.

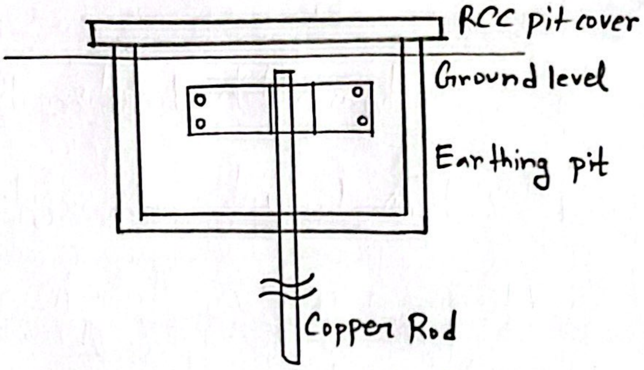

Fig 01: Copper Rod Earthing

(ii) Copper plate Earthing: A copper plate is buried horizontally in a pit providing a large surface area for effective fault current dissipation.

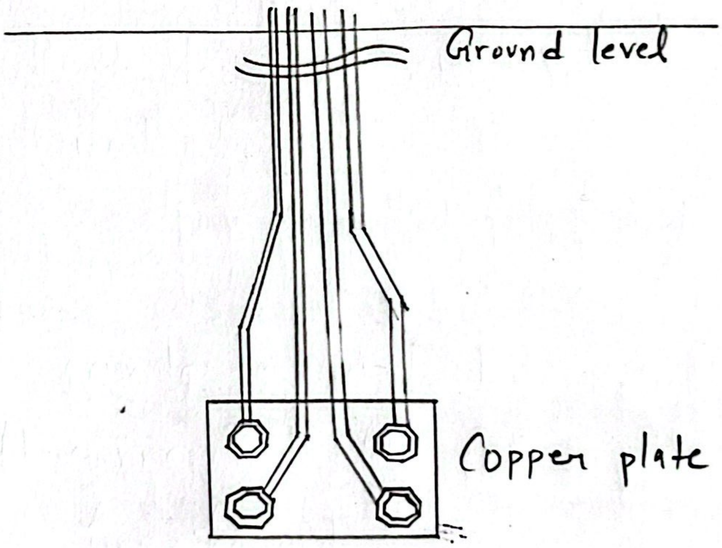

Fig 02: Copper Plate Earthing

(iii) Pipe Earthing: A GI pipe or perforated pipe is placed vertically in the soil to improve moisture retention and ground reliability.
It has the same configuration as Copper Rod

Earthing except just the copper Rod is replaced by a GI pipe.

Telecommunication Systems:

1) The wiring required for telecommunication in buildings includes placing concealed 2 pair indoor cables which are needed to get telephone lines of the wired telephone companies inside rooms and to get PABX lines of the buildings to the respective rooms under PABX. These 2 pair cables carry connection to 2 outlets:

(i) RJ11: Used for standard telephone line connections inside residential buildings

(ii) RJ45: Used for data and network services such as internet or LAN connections.

These cables ensure reliable signal transmission and reduce interference.

2) Television Antennas / Cable TV System :-

A minimum spacing of 350 mm must be ~~maintain~~ maintained to prevent electrical interference. This separation helps keep the TV and data signals stable and prevents noise caused by nearby power cables.

3) LAN networks: Cat-6 UTP cable is recommended for LAN systems because it supports higher data speeds and reduced crosstalk. It allows stable performance for modern-high BW internet connections.

Inspection and Testing:

1) Inspections are required before energizing the wiring. Before switching on power, the following must be checked:-

(i) Check Cable Color Coding to ensure proper phase, neutral and earth identification.

(ii) Check earthing continuity to confirm if earth path is connected throughout building.

(iii) Insulation resistance must be at least 5 MΩ to verify that wires are not leaking current.

2) The acceptable Meggar test values are:

(i) Circuits below 230 vols: 500 volts Meggar

(ii) Circuits between 230 to 400 volts: 1000 volts Meggar

(iii) The minimum acceptable insulation resistance value is 5 MΩ

(iv) Minimum earth continuity resistance 1 Ω

These values confirm that electrical system is properly insulated

Codes and Standards:

(i) BS 7671: British wiring regulations widely used for safe electrical design.

(ii) NEC (USA): Standard for electrical safety and grounding practices.

(iii) IEC Standards: International Standards ensuring compatibility and safety.

(iv) ISO 50001: Standard for energy management and efficiency in buildings.

(v) VDE (Verband Deutscher Elecktrotechniker):

Association of German Electrical Engineers.

These standards ensure uniformity and safety in building wiring.

Lightning protection:

The purpose of Lightning protection is to safely intercept lightning strikes and conduct the discharge current to the ground without damaging the structure.

1) The lightning protection methods:

(i) Franklin Rod System:- A simple metal rod installed at the highest point of a structure; it attracts lightning and safely directs the current to the ground.

(ii) Tight wire System:- Overhead wires are mounted around the structure to form a protective shield, diverting lightning strikes away from critical areas.

(iii) NFPA-780 Protection:- A standardized guideline that classifies protection levels and zones for structures, helping design lightning protection based on risk and geometry.

(iv) ESE ~~Air~~ Terminal:- An advanced rod that relases early streamers to capture lightning sooner, increasing the protected area compared to traditional rods.

(v) ABB OPR System:- A commercial early streamer emission device with precise design for high reliability protection in industrial or sensitive installations.

Each method creates a controlled path for lightning to safely reach the ground. This prevents fire, structural failure, equipment loss and electrical hazards caused by extremely high surge currents.

2) The Risk index is a numerical score based on environmental, structural and usage factors that determines the likelihood of lightning damage. A risk index value of 40 or higher indicates that lightning protection is necessary.

3) The example layouts observed throughout the experiment show:

(i) Position of air terminals on roof edges.

(ii) Paths of down conductors along the building.

(iii) Earthing pit placement and layout

(iv) Protection zone coverage around a building.

These diagrams help understand practical lightning protection design.

Discussion: This experiment covers essential practical electrical services in buildings. They explain earthing methods to safely dissipate fault currents; Structured telecommunication and data cabling to ensure reliable signals and safe separation from power lines and inspection or testing procedures to verify insulation and earth resistance before use. The section on lightning protection presents various systems, from traditional rod-based methods to advanced ESE air terminals, illustrated with example layouts. In the end, the content connects theoretical principles to practical, safety oriented electrical design.

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